

Skills Summary

Writing Inequalities from Verbal Constraints

Use this reference while completing your worksheet or when studying.

At Most and At Least: the Boundary Counts

Rule: “at most”, “no more than”, “maximum”, “up to” all *include* the limit, so they become $\leq$. “At least”, “no fewer than”, “minimum” include it too, and become $\geq$.

Test-value model: try the limit itself. If that value is allowed, the sign has the bar under it.

Example: A van may carry at most 900 kilograms, and each crate weighs 60 kilograms. Write and solve an inequality for the number c of crates.

Step 1. “At most” allows exactly 900 kilograms, so the sign is $\leq$ and the total mass of the crates is $60c$: $60c \leq 900$.

Step 2. Divide both sides by 60, which is positive so the direction does not change: $c \leq 15$, that is $c \leq 15$.

Try it: A test is passed with at least 45 points, and each correct question is worth 5 points. Write and solve an inequality for the number q of correct questions.

check: $6 \geq 6$

More Than and Fewer Than: the Boundary is Out

Rule: “more than”, “over”, “above” become $>$; “fewer than”, “less than”, “under”, “below” become $<$. The limit itself is *not* allowed, so the solution stops just short of it.

Test-value model: try the limit. If it is not allowed, the sign has no bar.

Example: Fewer than 30 cakes are packed into boxes of 5. Write and solve an inequality for the number x of boxes.

Step 1. x boxes of 5 cakes is $5x$ cakes, and “fewer than” excludes 30 itself, so the sign is strict: $5x < 30$.

Step 2. Divide both sides by 5: $x < 6$, that is $x < 6$ — six boxes would be exactly 30 cakes, which is one too many.

Try it: A hall has more than 48 seats, arranged in 8 equal rows. Write and solve an inequality for the number r of seats in one row.

check: $9 < 8$

Limits with a Starting Amount

Rule: when one amount is charged *once* and another repeats, the inequality has two terms: (rate) $\times$ (unknown) + (one-time amount), compared with the limit. The one-time amount is added, never multiplied by the count. Undo the addition first, then the multiplication.

Example: A phone plan costs 15 dollars a month plus 2 dollars for each gigabyte, and the budget is at most 27 dollars. Write and solve an inequality for the number g of gigabytes.

Step 1. The 2 dollars repeats for each gigabyte and the 15 dollars is paid once, and “at most” includes the limit: $2g + 15 \leq 27$.

Step 2. Subtract 15 from both sides to get $2g \leq 12$, then divide by 2: $g \leq 6$, that is **$g \leq 6$** .

Try it: A club charges 25 dollars to join plus 4 dollars a month, and a member has at most 65 dollars. Write and solve an inequality for the number m of months.

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(!) **Watch out:** an inequality has a whole *set* of solutions, and the situation may allow only some of them. “ $n \leq 15$ people” really means the whole numbers up to 15 — the algebra cannot know that people come in whole numbers.